Full Stack Assignment – 43

Assignment ID: FSD-43

Title: Version-Controlled Record Management System

Duration: 3 Hours

1️⃣ Objective

Design and develop a **Version-Controlled Record Management System** that maintains multiple versions of a record while ensuring only the latest version is editable.
The system must enforce version immutability rules and ensure **data persistence using a database**.

2️⃣ User Roles

EDITOR

- Register and log in to the system
- Create records
- Update records (creates a new version)
- View record version history

VIEWER

- Register and log in to the system
- View records and version history

3️⃣ Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** on APIs and frontend views

Record Management

- EDITOR can create records with basic fields
- Each update must create a **new version**
- All record versions must be stored in the database
- Version metadata (version number, timestamp) must be persisted

Version Handling

- Latest version is editable
- Previous versions are **read-only**

- Version history must be retrievable in order

4️⃣ Business Rule (Critical)

⚠️ **Previously created versions must not be modified once a new version is created.**

Only the latest version of a record is allowed to be edited.

5️⃣ Technical Constraints (Mandatory)

- All record and version data **must persist in a database**
- Hardcoded or in-memory version handling is **not allowed**
- Backend APIs must reflect actual database state
- JWT authentication and role validation are mandatory
- Use environment variables for sensitive configuration

6️⃣ Frontend Requirements

- Login and registration pages
- Role-based dashboards for EDITOR and VIEWER
- Record creation and update view for EDITOR
- Record listing with version history view
- Clear indication of latest vs historical versions

UI design is not evaluated; versioning logic correctness is the priority.

7️⃣ Submission Requirements (Strict)

You must submit **all of the following**:

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to a database

README.md (Mandatory)

The README file **must include**:

- Project overview

- Tech stack used
- User roles and permissions
- API endpoints
- Database schema / collections
- Steps to run locally
- Live frontend and backend links

8️⃣ Evaluation Note

Evaluation focuses on **version immutability, database persistence, role enforcement, and end-to-end execution**.